RISHAL RAJ MASKEY

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Assistant Network Administrator (L2) at Silverlining Data Center with an established background in 24/7 NOC operations, enterprise network monitoring, alert triage, incident response, and SLA-driven infrastructure delivery. Holds a BSc (Hons) in Computer Networking & IT Security. Hands-on experience across data center switching and routing, dual-WAN failover, clustered virtualization (3-node Proxmox VE), and telemetry pipelines (Grafana, InfluxDB, SNMP v3).

PROFESSIONAL EXPERIENCE

Assistant Network Administrator — L2 · Silverlining Data Center, Kathmandu **Effective Aug 2026 – Present**

- Promoted from NOC operations to L2 Network Administration, taking operational responsibility for data center switching, routing, and network uptime.
- Handle L2 network escalations, VLAN configurations, and incident resolution for enterprise and banking-sector clients.
- Perform proactive network health audits, link failover assessments, and infrastructure maintenance across high-density data center environments.

NOC Engineer · Silverlining Data Center, Kathmandu **2024 – Jul 2026**

- Delivered 24/7 network and systems monitoring, alert triage, and rapid incident response in a live data center environment.
- Orchestrated monthly ITSM reporting (incidents, service requests, changes, problems) via Freshservice for internal teams and corporate clients.
- Prepared and delivered daily mission-critical data replication status reports and monthly performance analytics for key clients, including the banking sector.
- Coordinated multi-tier escalations across network engineering, systems teams, and security tooling alerts (EDR & SIEM).

L1 Support · Vianet Communications, Lalitpur **Jul 2023 – Oct 2023**

- Provided frontline technical assistance and diagnostics for residential and enterprise internet subscribers.
- Configured customer-premises equipment (CPE routers, switches, ONTs) and resolved escalated physical/data-link network impairments.

HOMELAB INFRASTRUCTURE (3-NODE PROXMOX CLUSTER)

A production-grade personal lab mirroring enterprise data center topologies for hands-on virtualization, routing, and observability:

- **Clustered Compute:** 3-node Proxmox VE high-availability cluster running distributed workloads: TrueNAS SCALE VM (ZFS storage, NFS/SMB shares, app containers), dedicated Ubuntu monitoring server, and staging VMs across Linux and Windows Server.
- **Dual-WAN & Zero-Trust Networking:** TP-Link ER605 edge router balancing primary Worldlink and backup CG Net lines with automated failover. Remote access orchestrated via WireGuard (dedicated static public IP on WAN2), Cloudflare Tunnel, and Tailscale mesh, fronted by Nginx Proxy Manager with wildcard SSL certificates (`*.maskey.lab`).
- **Observability & Telemetry Stack:** Router SNMP v3 metrics pulled via Telegraf into InfluxDB and visualized on real-time Grafana dashboards; Beszel agents for per-node compute metrics; Speedtest Tracker monitoring dual-ISP throughput and latency.
- **Self-Hosted Services:** Immich (photos), Nextcloud (files), Emby (media), Calibre (books), AdGuard Home (DNS filtering & local rewrites), and Homarr (centralized services dashboard).

FEATURED TECHNICAL PROJECTS

NEPSE Whale Desk & AI Market Analytics Dashboard **2025 – 2026**

Tech: Python (FastAPI), React, Vite, Tailwind CSS, SQLite/SQLAlchemy, AI/LLM API Integration, Web Scraping

- Built a full-stack real-time analytics platform and market intelligence dashboard for the Nepal Stock Exchange (NEPSE).
- Implemented live data ingestion tracking market breadth, 10% circuit limits, >3x institutional volume spikes, and multi-broker floorsheet accumulation/distribution patterns.
- Integrated an on-demand AI strategy engine powered by LLMs to synthesize technical momentum, volume anomalies, and broker activity into actionable trade suggestions and risk assessments.
- Architected responsive Bloomberg-inspired dark-mode UI with candlestick charts, sector heatmaps, and portfolio mark-to-market P/L tracking.

Smart Traffic Management for Emergency Vehicles (STEM) **Final Year Capstone · 2024**

Tech: Python, OpenCV, Machine Learning, Computer Vision, Web Interface, IoT Simulation

- Developed an intelligent traffic preemption system that dynamically clears green corridors for approaching ambulances and emergency vehicles.
- Engineered optical signal detection and distance-threshold triggers using computer vision models to automate signal switching at intersections, significantly reducing transit delays.

EDUCATION

BSc (Hons) Computer Networking & IT Security · Islington College, Kathmandu **Oct 2021 – Dec 2024**

In partnership with London Metropolitan University, UK

- Core studies: Network Architecture, Routing & Switching, Network Security, Ethical Hacking, Linux/Unix Administration, Server Virtualization, and Systems Defense.

High School Degree (Science / Computer Science) · Uniglobe Secondary School, Kathmandu **Jul 2019 – Jun 2021**

TRAINING & CERTIFICATIONS

AWS Academy Graduate · Amazon Web Services **2023**

- **AWS Academy Graduate — Data Center Technician:** Data center infrastructure, power distribution, server maintenance, cabling, and cooling systems.
- **AWS Academy Graduate — Cloud Fundamentals & Security:** Core AWS cloud infrastructure, IAM access controls, network security policies, and compliance.
- **AWS Academy Graduate — Cloud Architect:** High-availability architecture, VPC peering, subnets, and scalable cloud service design.

TECHNICAL SKILLS

Networking & Security: Switching & routing, Dual-WAN load balancing & failover, VLANs, WireGuard, Tailscale, Cloudflare Tunnel, SNMP v3, DNS/DHCP, EDR alert triage

Virtualization & OS: Proxmox VE (3-Node Cluster, LXC/KVM), TrueNAS SCALE (ZFS, NFS/SMB), Linux (Ubuntu, Debian), Windows Server, Active Directory, VMware ESXi

Monitoring & ITSM: Grafana, InfluxDB, Telegraf, Beszel, Freshservice (ITSM), Daily replication status reporting, SLA tracking, 24/7 alert triage

Software & Tools: Python (FastAPI, Web Scraping, Scripting), SQLite/SQLAlchemy, HTML5, CSS3, JavaScript (React), Docker containers, Nginx Reverse Proxy, Git/GitHub

Languages: Nepali (Native), English (Fluent / Professional)